

Web Programming

Web Programming

Rodrigo Martins Pagliares

Web programming foundations

**Writing, deploying and
accessing websites**

In this **module** we are going **write** a
simple **example** with **HTML**, **CSS**, and
JavaScript

We are going to demonstrate several tools for deploying and accessing the example.

Minimum software needed

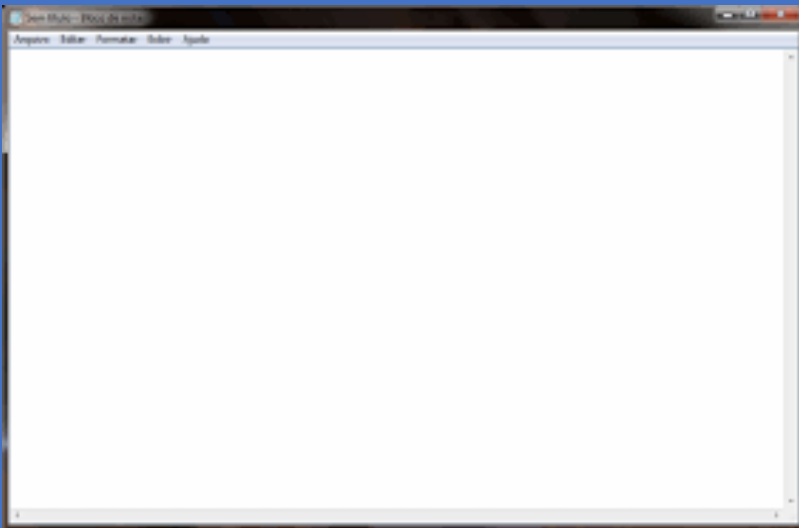
Lecture: Minimum software needed

Module: Writing, deploying and accessing websites

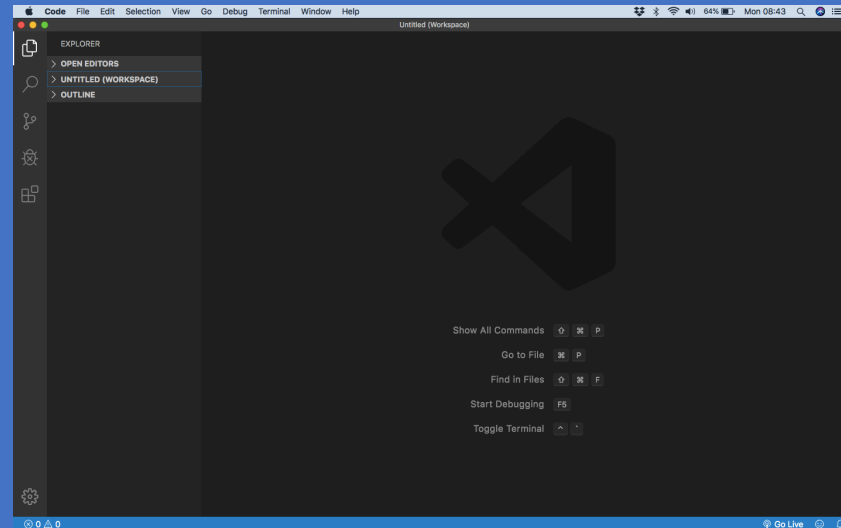
**Coding with HTML, CSS, and
JavaScript demands a basic
text/code editor that can handle
raw TXT files.**

Minimum software needed

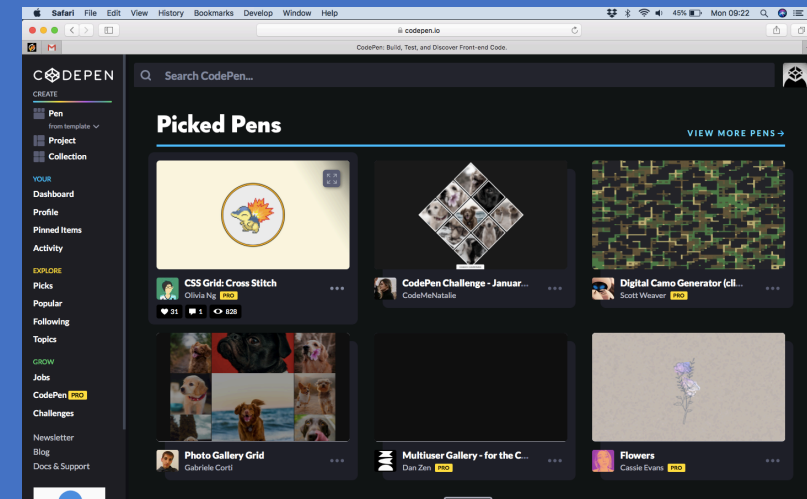
1. A text editor like **TextEdit** or **Notepad** or a code editor like **Visual Studio Code**.
 - ✓ Alternatively, you can use an online text editor such as **CodePen** or **JSBin**.



Notepad



Visual Studio Code

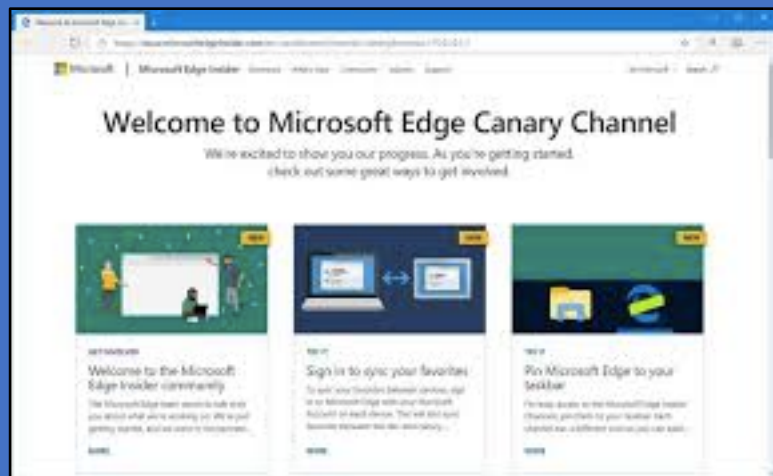


CodePen

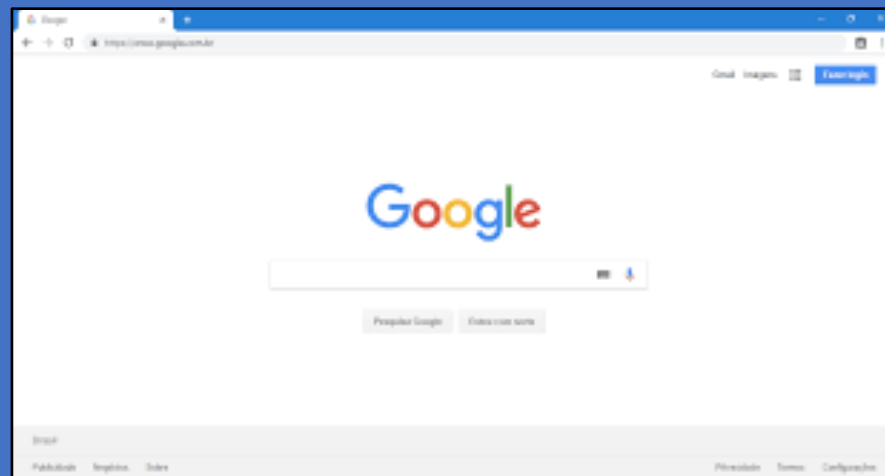
Minimum software needed

2. A web browser like Google Chrome, Mozilla Firefox, or Apple's Safari.

- ✓ Running at the same time with the text/code editor, often side by side.
- ✓ Often with the browser developer tools opened.



Microsoft Edge



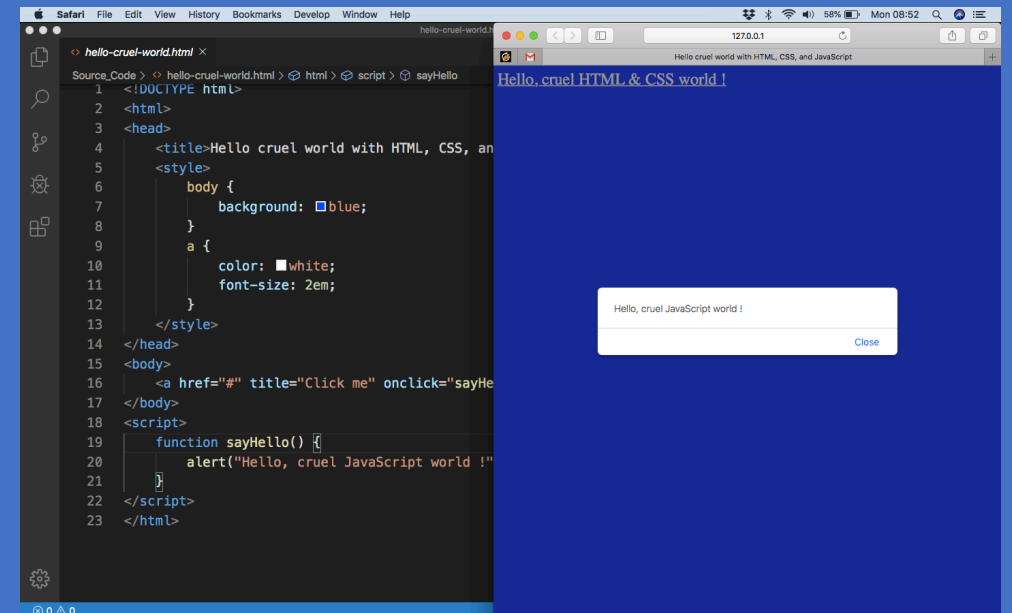
Google Chrome



Mozilla Firefox

Minimum software needed

- As you **develop** your **project**, you **work** on the **code** in both **code editor** and **web browser**.
 - ✓ Sometimes writing code in the **code editor**.
 - ✓ Sometimes either **interacting** with the **live site** in the **browser**.
 - ✓ Other times **manipulating** and **editing** the **code directly** in the **browser developer tools**.



Hello, cruel world in HTML, CSS, and JavaScript

Lecture: Hello, cruel world in HTML, CSS, and JavaScript

Module: Writing, deploying and accessing websites

Hello, cruel HTML, CSS, and JavaScript world example

```
<!DOCTYPE html>
<html>
<head>
  <title>Hello cruel world with HTML, CSS, and JavaScript</title>
  <style>
    body {
      background: blue;
    }
    a {
      color: white;
      font-size: 2em;
    }
  </style>
</head>
<body>
  <a href="#" title="Click me" onclick="sayHello()">Hello, cruel HTML & CSS world HTML !</a>
</body>
<script>
  function sayHello() {
    alert("Hello, cruel JavaScript world !");
  }
</script>
```

HTML

CSS

JavaScript

web-programming-foundations/hello-cruel-world.html

Hello, cruel HTML, CSS, and JavaScript world example

```
<!DOCTYPE html>
<html>
<head>
  <title>Hello cruel world with HTML, CSS, and JavaScript</title>
  <style>
    body {
      background: blue;
    }
    a {
      color: white;
      font-size: 2em;
    }
  </style>
</head>
<body>
  <a href="#" title="Click me" onclick="sayHello()">Hello, cruel HTML & CSS world HTML !</a>
</body>
<script>
  function sayHello() {
    alert("Hello, cruel JavaScript world !");
  }
</script>
</html>
```

HTML

CSS

JavaScript

HTML

Notice we internalize the CSS rules and JavaScript code within HTML. This is a BAD practice and is used here to show that it is possible, but this should be avoided by externalizing CSS rules and JavaScript code.

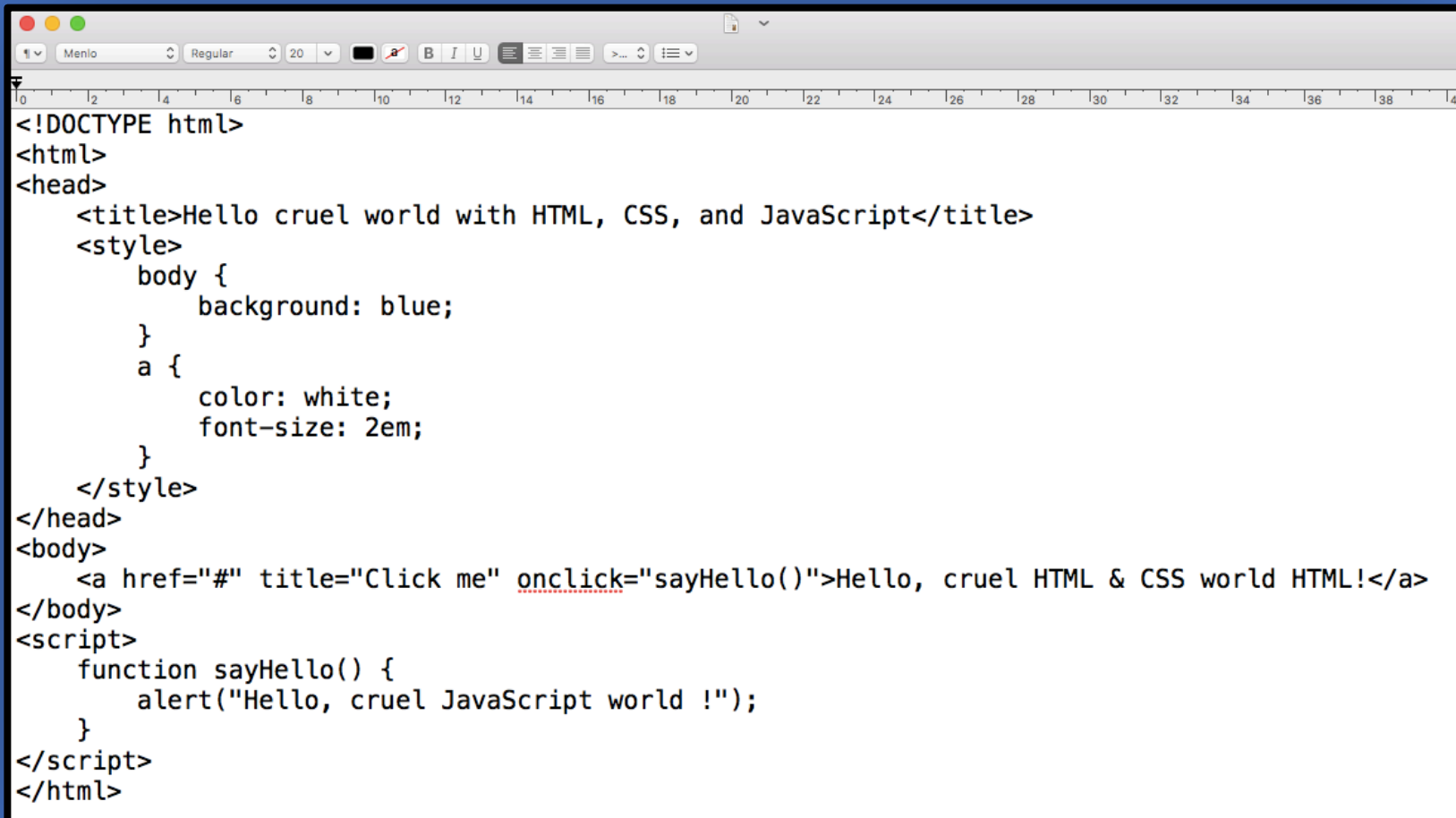


Writing the example in a text editor

Lecture: Writing the example in a text editor

Module: Writing, deploying and accessing websites

Text editors

A screenshot of a Mac OS TextEdit window. The window has a title bar with red, yellow, and green window control buttons. Below the title bar is a menu bar with 'Menlo' selected in the font menu, and 'Regular' and '20' in the style and size menus. The main text area contains HTML and JavaScript code. The code is as follows:

```
<!DOCTYPE html>
<html>
<head>
  <title>Hello cruel world with HTML, CSS, and JavaScript</title>
  <style>
    body {
      background: blue;
    }
    a {
      color: white;
      font-size: 2em;
    }
  </style>
</head>
<body>
  <a href="#" title="Click me" onclick="sayHello()">Hello, cruel HTML & CSS world HTML!</a>
</body>
<script>
  function sayHello() {
    alert("Hello, cruel JavaScript world !");
  }
</script>
</html>
```

Mac OS

- **TextEdit**

Windows

- **Notepad**

[Mac](#)[iPad](#)[iPhone](#)[Watch](#)[TV](#)[Music](#)[Support](#)

TextEdit User Guide

[Communities](#)[Contact Support](#)

Select version:

macOS Big Sur 11.0



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Work with HTML documents in TextEdit on Mac

You can use TextEdit to edit or display HTML documents as you'd see them in a browser (images may not appear), or in code-editing mode.

Note: By default, curly quotes and em dashes are substituted for straight quotes and hyphens when editing HTML as formatted text. (Code-editing mode uses straight quotes and hyphens.) To learn how to change this preference, see [New Document options](#).

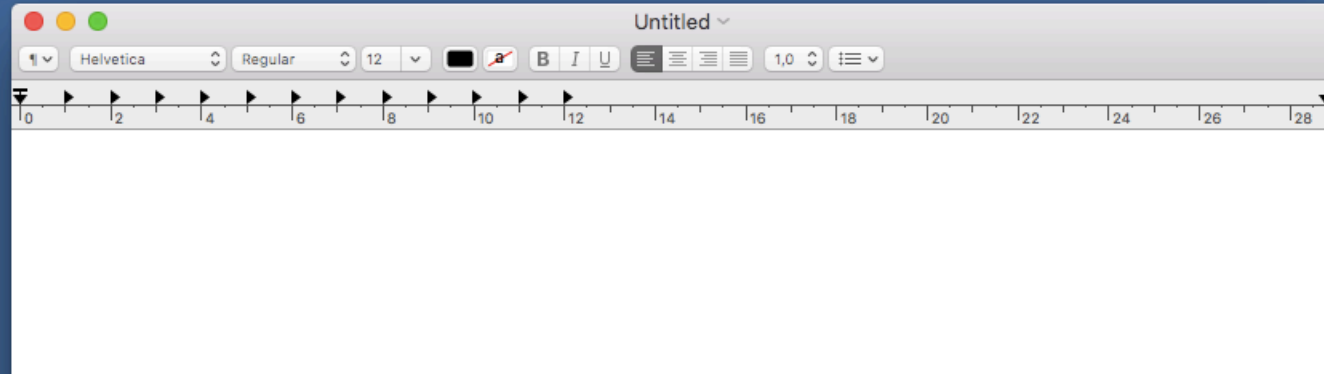
Create an HTML file

1. In the TextEdit app on your Mac, choose File > New, then choose Format > Make Plain Text.
2. Enter the HTML code.
3. Choose File > Save, type a name followed by the extension .html (for example, enter index.html), then click Save.

on to use, click "Use .html."

<https://apple.co/36wTPd3>

View an HTML document



We must assure our **text editor** is treating the **file** as a **raw text file**.

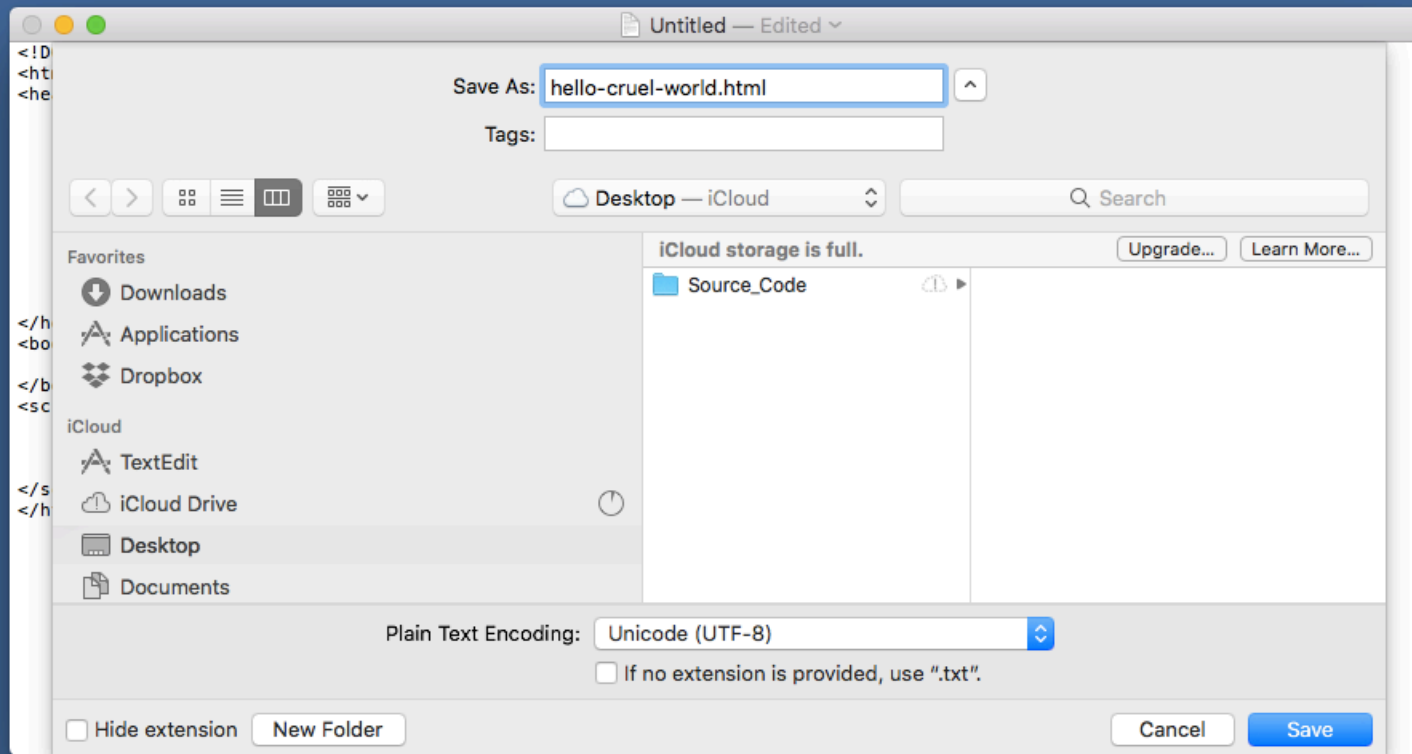


Mac OS TextEdit



```
Untitled — Edited v
Menlo Regular 16
<!DOCTYPE html>
<html>
<head>
  <title>Hello cruel world with HTML, CSS, and JavaScript</title>
  <style>
    body {
      background: blue;
    }
    a {
      color: white;
      font-size: 2em;
    }
  </style>
</head>
<body>
  <a href="#" title="Click me" onclick="sayHello()">Hello, cruel HTML & CSS world !</a>
</body>
<script>
  function sayHello() {
    alert("Hello, cruel JavaScript world !");
  }
</script>
</html>
```

Type in the web document



Save the file

```
hello-cruel-world.html
Menlo Regular 16
<!DOCTYPE html>
<html>
<head>
  <title>Hello cruel world with HTML, CSS, and JavaScript</title>
  <style>
    body {
      background: blue;
    }
    a {
      color: white;
      font-size: 2em;
    }
  </style>
</head>
<body>
  <a href="#" title="Click me" onclick="sayHello()">Hello, cruel HTML & CSS world !</a>
</body>
<script>
  function sayHello() {
    alert("Hello, cruel JavaScript world !");
  }
</script>
</html>
```



Source_Code



hello-cruel-world.html

```
hello-cruel-world.html — Edited
<!DOCTYPE html>
<html>
<head>
  <title>Hello cruel world with HTML, CSS, and JavaScript</title>
  <style>
    body {
      background: blue;
    }
    a {
      color: white;
      font-size: 2em;
    }
  </style>
</head>
<body>
  <a href="#" title="Click me" onclick="sayHello()">Hello, cruel HTML & CSS world !</a>
</body>
<script>
  function sayHello() {
    alert("Hello, cruel JavaScript world !");
  }
</script>
</html>
```

Source_Code

hello-cruel-world.html

Accessing the example in a web browser

Lecture: Accessing the example in a web browser
Module: Writing, deploying and accessing websites

```
hello-cruel-world.html — Edited
<!DOCTYPE html>
<html>
<head>
  <title>Hello cruel world with HTML, CSS, and JavaScript</title>
  <style>
    body {
      background: blue;
    }
    a {
      color: white;
      font-size: 2em;
    }
  </style>
</head>
<body>
  <a href="#" title="Click me" onclick="sayHello()">Hello, cruel HTML & CSS world !</a>
</body>
<script>
  function sayHello() {
    alert("Hello, cruel JavaScript world !");
  }
</script>
</html>
```



Source_Code



hello-cruel-world.html



The out-of-the-box way of testing our web documents is to run the files natively from the file system in the browser.

```
hello-cruel-world.html — Edited
<!DOCTYPE html>
<html>
<head>
  <title>Hello cruel world with HTML, CSS, and JavaScript</title>
  <style>
    body {
      background: blue;
    }
    a {
      color: white;
      font-size: 2em;
    }
  </style>
</head>
<body>
  <a href="#" title="Click me" onclick="sayHello()">Hello, cruel HTML & CSS world !</a>
</body>
<script>
  function sayHello() {
    alert("Hello, cruel JavaScript world !");
  }
</script>
</html>
```

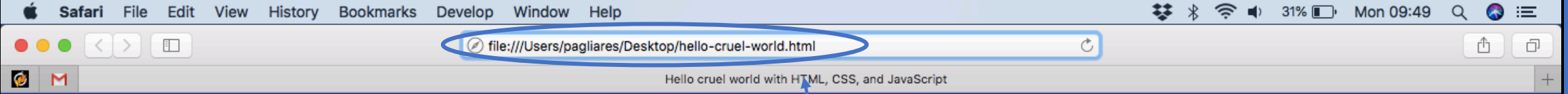


Source_Code



hello-cruel-world.html

Double-click to run the HTML document



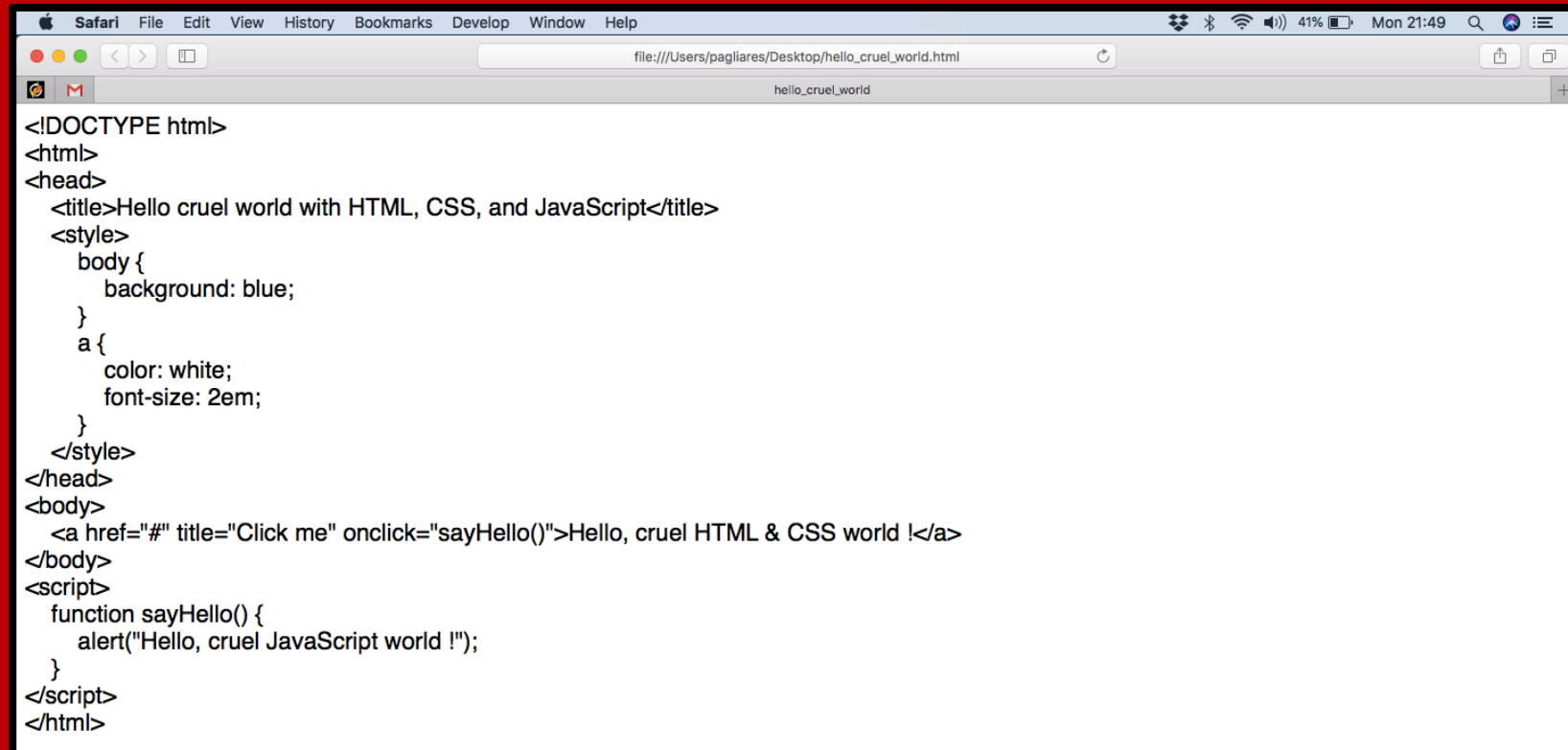
Hello, cruel HTML & CSS world !

The URL shows the path to the document
in our local system

Hello, cruel JavaScript world !

Close

Problem !

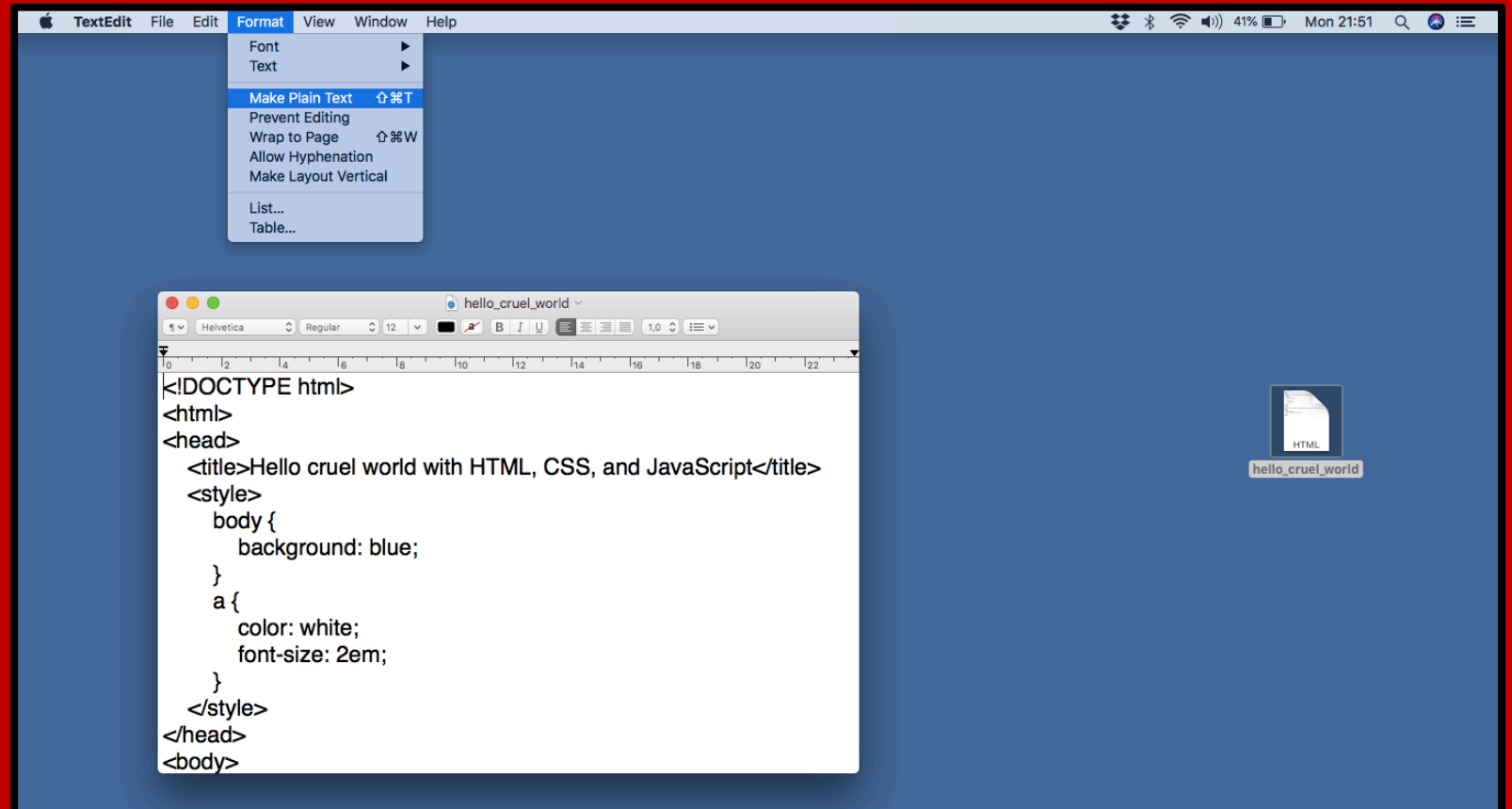


The screenshot shows a Safari browser window with the address bar displaying 'file:///Users/pagliares/Desktop/hello_cruel_world.html'. The page title is 'hello_cruel_world'. The source code is displayed in the main content area, showing an HTML document with a blue background and a white link that says 'Hello, cruel HTML & CSS world !'. The link has an onclick event that calls a JavaScript function 'sayHello()', which displays an alert box with the message 'Hello, cruel JavaScript world !'.

```
<!DOCTYPE html>
<html>
<head>
  <title>Hello cruel world with HTML, CSS, and JavaScript</title>
  <style>
    body {
      background: blue;
    }
    a {
      color: white;
      font-size: 2em;
    }
  </style>
</head>
<body>
  <a href="#" title="Click me" onclick="sayHello()">Hello, cruel HTML & CSS world !</a>
</body>
<script>
  function sayHello() {
    alert("Hello, cruel JavaScript world !");
  }
</script>
</html>
```

If you see an output like this instead of the one shown in the previous slide....

Troubleshooting !



Make sure to save your file in a raw txt format (plain text)

Accessing a webpage without a webserver

- Drawbacks:

- ✓ Manually reload the browser every time we make a change in the editor.
- ✓ Not the same as running an HTML document from a web server.
 - We miss a bunch of the web functionality and many features will not work.



Source: <https://bit.ly/3ywRFeU>

Writing the example in online code editors

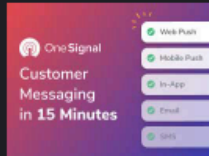
Lecture: Writing the example in online code editors
Module: Writing, deploying and accessing websites

CodePen

[Start Coding](#)

Trending
Challenges
Spark

CodePen **PRO**



The market leader in customer messaging, powering push notifications, email, SMS & in-app messages.
(ads via Carbon)



The best place to build, test, and discover front-end code.

CodePen is a **social development environment** for front-end designers and developers. Build and deploy a website, show off your work, build test cases to learn and debug, and find inspiration.

[Sign Up for Free](#)

Build & Test

Get work done quicker by building out entire projects or isolating code to test features and animations. Want to keep it all under wraps? Upgrade to a **PRO** account to keep your work private.

[Try the Editor](#)

Learn & Discover

Just starting out? Browsing, forking, and playing with Pens is a great way to understand how they were built and how code works. Teaching a class? **Professor mode** lets your students to see and comment on real-time Pen updates.

[Learn More](#)

Share Your Work

Become a part of the most active front-end community in the world by sharing work.

<https://codepen.io>

HTML

```
<div class="rect"></div>
```

SCSS

```
.rect {  
  t
```

JS

```
var colors =  
["#74B087", "#DE7300", "#74B087"];  
  
function animate() {};
```



HTML

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>Hello cruel world with HTML, CSS, and JavaScript</title>
5   <style>
6     body {
7       background: blue;
8     }
9     a {
10      color: white;
11      font-size: 2em;
12    }
13  </style>
14 </head>
15 <body>
16   <a href="#" title="Click me" onclick="sayHello()">Hello, cruel HTML & CSS world !
17   </a>
18 </body>
19 <script>
20   function sayHello() {
21     alert("Hello, cruel JavaScript world !");
22   }
23 </script>
24 </html>
```

Hello, cruel HTML & CSS world !

<https://codepen.io/pagliares/pen/yLyQGyQ>



Delete

Add to Collection

Fork

Embed

Export

Share



HTML

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>Hello cruel world with HTML. CSS. and JavaScript</title>
```

Hello, cruel HTML & CSS world !

When using Codepen, we just type the **HTML** contents of the **body** element. That's the reason of this warning.

```
12 }
13 </style>
14 </head>
15 <body>
16 <a href="#" title="Click me" onclick="sayHello()">Hello, cruel HTML & CSS world !
17 </a>
18 </body>
19 <script>
20 function sayHello() {
21   alert("Hello, cruel JavaScript world !");
22 }
23 </script>
24 </html>
```



CSS

7 unsaved changes X

JS





hello_cruel_world_html_css_javascript

Rodrigo Martins Pagliares



Save

Settings

Change View



HTML

22 unsaved changes X

```
1
2 <a href="#" title="Click me" onclick="sayHello()">Hello, cruel HTML & CSS world !</a>
3
```

Hello, cruel HTML & CSS world !

CSS

```
1 body {
2   background: blue;
3 }
4 a {
5   color: white;
6   font-size: 2em;
7 }
8
```

HTML, CSS and JavaScript are on their respective panels now. Notice that in Codepen we do not need to declare the HTML elements `<body>`, `<style>` and `<script>`

JS

```
1 function sayHello() {
2   alert("Hello, cruel JavaScript world !");
3 }
4
```

Console

Assets

Comments

⌘ Keys



CodePen: Unlock all of CodePen



Last saved OVER 1 YEAR AGO



Delete

Add to Collection

Fork

Embed

Export

Share

HTML

22 unsaved changes X

```
1  
2 <a href="#" title="Click me" onclick="sayHello()">Hello, cruel HTML & CSS world !</a>  
3
```

CSS

```
1 body {  
2   background: blue;  
3 }  
4 a {  
5   color: white;  
6   font-size: 2em;  
7 }  
8
```

JS

```
1 function sayHello() {  
2   alert("Hello, cruel JavaScript world !");  
3 }  
4
```

Hello, cruel HTML & CSS world !

You can try codepen without creating an account. Users with a free account can save their web documents. Users who pay for the service can use several additional features.



HTML

14 unsaved changes X



```
1
2 <a href="#" title="Click me" onclick="sayHello()">Hello, cruel HTML & CSS world !</a>
3
```

Hello, cruel HTML & CSS world !

CSS



```
1 body {
2   background: blue;
3 }
```

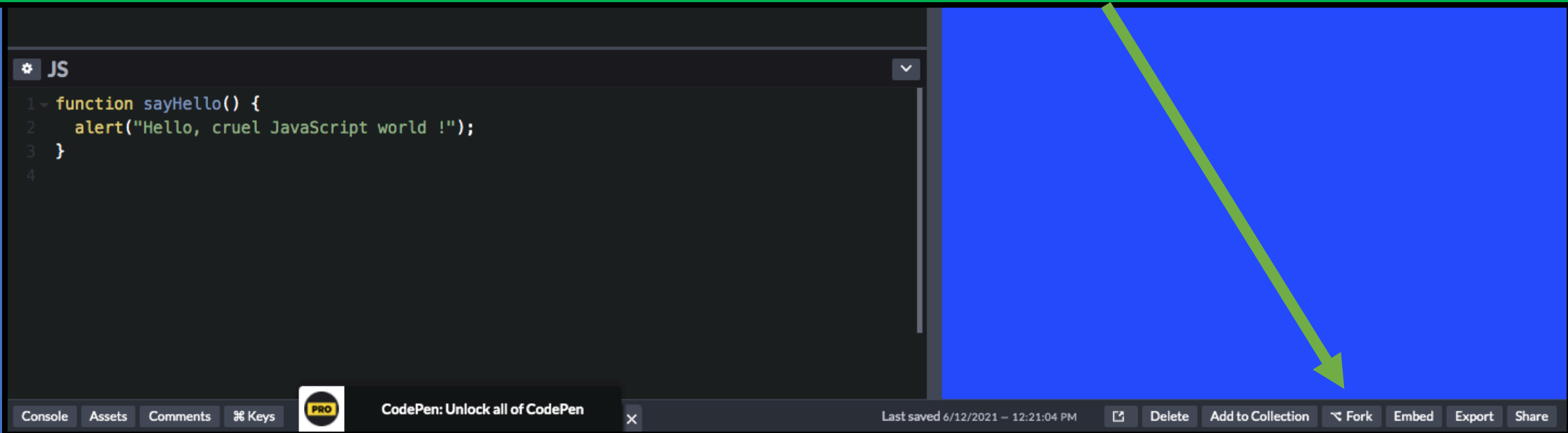
You can experiment with this example by editing it and observing the results. No changes are saved and you lose the changes after reloading the web browser.

```
2 alert("Hello, cruel JavaScript world !");
3 }
4
```





You can with a **free account**, “**copy**” some of the **thousands** of **projects** available on **CodePen** to your **personal account**, **enabling** them to be **reused** in your **own projects**. Select the option **Fork** and **check** your **work** in the **main page** of **CodePen**.





CREATE

Pen

Project

Collection

Your Work

Activity

Assets

Pinned Items

Following

Trending

Challenges

Spark

CodePen PRO

Following

Trending

Your Work



Picked by CodePen

Tab 1 Tab 2 Tab 3

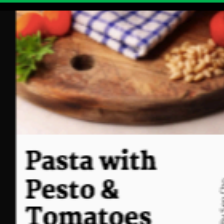
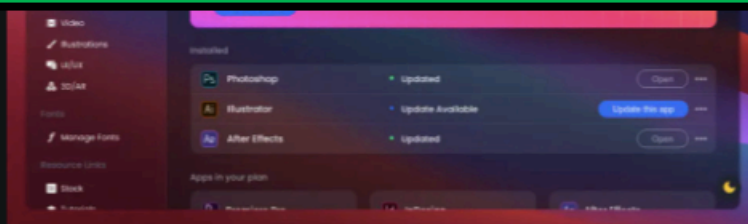
Tab 1

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut tempus, velit et tempor molestie, urna purus. In magna neque, nec condimentum purus. Nisi eu tortor. Curabitur non maximus mi. Suspendisse cursus libero nisi, ac imperdiet purus malesuada vel. Cras maximus in quam vel sagittis. Sed ut cursus turpis. Mauris a ipsum congue, sagittis mius commodo, tincidunt mi. Sed mi ante, viverra et aculis sollicitudin, malesuada vitae turpis. Proin nisi neque, tempus a nisi et, dictum feugiat nulla.

Present sit amet molestie nunc, in nec tellus cursus, malesuada urna quis, tristique nisl. Vivamus laevis tempus libero. Mauris nec est tempus, volutpat dolor eu, gravida mauris. Maecenas sed ullamcorper justo. Cras sit amet massa augue. Nam posuere nec justo at nunc.

Pure CSS Tabs
mrwolverincCSS mouse-out transition effect
Adam Argyle PROCandy Color B
Yuhomyan

From the homepage, click **Trending** to see some examples of projects freely available.





CREATE

Pen

Project

Collection

Your Work

Activity

Assets

Pinned Items

Following

Trending

Challenges

Spark

CodePen PRO

Following

Trending

Your Work

Picked by CodePen

Tab 1 Tab 2 Tab 3

Tab 1

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut tempus, velit et tempus molestie, urna purus. Inaugula neque, nec condimentum purus. nibh eu tortor. Curabitur non maximus mi. Suspendisse cursus libero nisi, ac imperdiet purus malesuada vel. Cras maximus in quam vel sagittis. Sed ut cursus turpis. Mauris a ipsum congue, sagittis mius commodo, tincidunt mi. Sed mi ante, viverra et aculis sollicitudin, malesuada vitae turpis. Proin nisi neque, tempus a nisl et, dictum feugiat nulla.

Present sit amet molestie nunc, in nec tellus cursus, malesuada urna quis, tristique nisl. Vivamus laeulis tempus libero. Mauris nec elit tempus, volutpat dolor eu, gravida mauris. Maecenas sed ullamcorper justo. Cras sit amet massa augue. Nam posuere nec justo at nuctor.



Pure CSS Tabs

mrwolverinc



CSS mouse-out transition effect

Adam Argyle PRO

Hover Me

Picked by CodePen



Pasta with Pesto & Tomatoes

Select a **project** of your choice. Let's this *"Hover Me"* project. You can pick any other.

Pasta with Pesto & Tomatoes

Recipe Layout #CodePenChallenge

Candy Color B

Yuhomyan

Picked by CodePen

CSS Art: Men



HTML

```
1 <h1>Hover Me</h1>
```

CSS

```
21
22 h1 {
23   position: relative;
24   font-size: 5rem;
25 }
26
27 html {
28   block-size: 100%;
29   inline-size: 100%;
30 }
31
32 body {
33   min-block-size: 100%;
34   min-inline-size: 100%;
35   margin: 0;
36   box-sizing: border-box;
37   display: grid;
38   place-content: center;
39   font-family: system-ui, sans-
    serif;
40 }
41
42 @media (orientation: landscape) {
43   body {
44     grid-auto-flow: column;
45   }
46 }
47
```

JS (Babel)



Hover Me

HTML

```
1 <h1>Hover Me</h1>
```

CSS

```
21
22 h1 {
23   position: relative;
24   font-size: 5rem;
25 }
26
27 html {
28   block-size: 100%;
29   inline-size: 100%;
30 }
31
32 body {
33   min-block-size: 100%;
34   min-inline-size: 100%;
35   margin: 0;
36   box-sizing: border-box;
37   display: grid;
38   place-content: center;
39   font-family: system-ui, sans-
  serif;
40 }
41
42 @media (orientation: landscape) {
43   body {
44     grid-auto-flow: column;
45   }
46 }
47
```

JS (Babel)

The *Hover Me* text is highlighted with a blue background when we hover the mouse over the text.

Hover Me

HTML

```
1 <h1>Hover Me</h1>
```

CSS

```
22 h1 {  
23   position: relative;  
24   font-size: 2em;  
25 }  
26  
27 html {  
28   block-size: 100%;  
29   inline-size: 100%;  
30 }  
31  
32 body {  
33   min-block-size: 100%;  
34   min-inline-size: 100%;  
35   margin: 0;  
36   box-sizing: border-box;  
37   display: grid;  
38   place-content: center;  
39   font-family: system-ui, sans-  
   serif;  
40 }  
41  
42 @media (orientation: landscape) {  
43   body {  
44     grid-auto-flow: column;  
45   }  
46 }  
47
```

JS (Babel)

You can interact with the projects by editing the HTML, CSS and JavaScript code, but you are not able to save your changes. Here, I changed from 5em to 2em the font-size of the Element <h1>

Hover Me



HTML

```
1 <h1>Hover Me</h1>
```

CSS

```
21
22 h1 {
23   position: relative;
24   font-size: 5rem;
25 }
26
27 html {
28   block-size: 100%;
29   inline-size: 100%;
30 }
31
32 body {
33   min-block-size: 100%;
34   min-inline-size: 100%;
35   margin: 0;
36   box-sizing: border-box;
37   display: grid;
38   place-content: center;
39   font-family: system-ui, serif;
40 }
41
42 @media (orientation: landscape) {
43   body {
44     grid-auto-flow: column;
45   }
46 }
47
```

Hover Me

If you want to save the changes and reuse the project, you need to copy (fork) it to your account.

JS (Babel)





HTML

```
1 <h1>Hover Me</h1>
```

CSS

```
21
22 h1 {
23   position: relative;
24   font-size: 51m;
25 }
26
27 html {
28   block-size: 100%;
29   inline-size: 100%;
30 }
31
32 body {
33   min-block-size: 100%;
34   min-inline-size: 100%;
35   margin: 0;
36   box-sizing: border-box;
37   display: grid;
38   place-content: center;
39   font-family: system-ui, sans-serif;
40 }
41
42 @media (orientation: landscape) {
43   body {
44     grid-auto-flow: column;
45   }
46 }
47
```

Hover Me

After clicking fork, let's return the homepage.

JS (Babel)



CODEPEN

CREATE

Pen

Project

Collection

Your Work

Activity

Assets

Pinned Items

Following

Trending

Challenges

Spark

CodePen PRO

Search CodePen...

Following

Trending

Your Work

Pens

Projects

Collections

Deleted

SEARCH

Search for...

Search

VIEW

All

GRID

LIST

SORT BY

Date Created

SORT DIRECTION

TAG

Keyword...

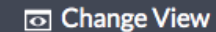
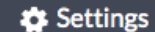
Title	Created	Last Updated	Stats			
CSS mouse-out transition effect	June 12, 2021	June 12, 2021	0	0	0	
html-dom-relationship_with_style	June 4, 2021	June 4, 2021	0	0	7	
lecture-css-first-steps	August 18, 2020	August 18, 2020	0	0	0	
hello_cruel_world_html_css_javascript	January 20, 2020	June 12, 2021	0	0	25	

Select your work. The project was copied to your own documents. Now you can reuse, edit and Save it.

CODEPEN

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Demo or it didn't happen.
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Community Spark Challenges Topics Code of Conduct



HTML

```
1 <h1>Hover Me</h1>
```

CSS

```
1 h1::before {  
2   transform: scaleX(0);  
3   transform-origin: bottom right;  
4 }  
5  
6 h1: hover::before {  
7   transform: scaleX(1);  
8   transform-origin: bottom left;  
9 }
```

JS (Babel)

```
1
```

Now you can reuse, edit and Save it.

Hover Me

hello_cruel_world_html_css_javascript
Rodrigo Martins Pagliares

Save Settings Change View

```
HTML 13 unsaved changes X
1 <a href="#" title="Click me" onclick="sayHello()">Hello, cruel HTML
  & CSS world !</a>
2

CSS
1 body {
2   background: blue;
3 }
4
5 a {
6   color: white;
```

Hello, cruel HTML & CSS world !

Codepen is a great way to create simple projects, share with others and make them accessible on the web. One drawback is the weird URLs you need to use to access the documents.

```
2 alert("Hello, cruel JavaScript world !");
3 }
4
```

JSBin

Not Secure — jsbin.com

JS Bin - Collaborative JavaScript Debugging



New bin

JS Bin features »

Getting started

Keyboard Shortcuts

Exporting/importing gist

☐ Textarea editor mode

Pro features »

Private bins

Dropbox backup

Vanity URLs

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The Return and The Refactor

Help »

Keyboard shortcuts

Gists: how to export to a gist

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By using JS Bin you agree to our legal terms

"Everyone should learn how to program a computer because it teaches you how to think" — Steve Jobs

File ▾ Add library

HTML CSS JavaScript Console Output

Login or Register Blog ¹ Help

HTML ▾

```
<!DOCTYPE html>
<html>
<head>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width">
  <title>JS Bin</title>
</head>
<body>

</body>
</html>
```

Output

Run with JS Auto-run JS ☒

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ethical ads by CodeFund • Close

https://jsbin.com

Apple Safari File Edit View History Bookmarks Develop Window Help

jsbin.com

JS Bin - Collaborative JavaScript Debugging

 [JS Bin features »](#) [Pro features »](#) [Blog »](#) [Donate to JS Bin ♥ »](#) "Everyone should learn how to program a computer because it teaches you how to think" — Steve Jobs

[Getting started](#) [Private bins](#) [The Return and The Refactor](#) [Support JS Bin to keep the project open source & MIT for all](#)

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☐ Textarea editor mode [Upgrade to pro now](#)

File Add library HTML CSS JavaScript Console Output Login or Register Blog ¹ Help

HTML

```
<!DOCTYPE html>
<html>
<head>
  <title>Hello cruel world with HTML, CSS, and JavaScript</title>
  <style>
    body {
      background: blue;
    }
    a {
      color: white;
      font-size: 2em;
    }
  </style>
</head>
<body>
  <a href="#" title="Click me" onclick="sayHello()">Hello, cruel HTML & CSS world !
</body>
<script>
  function sayHello() {
    alert("Hello, cruel JavaScript world !");
  }
</script>
</html>
```

Output

Run with JS Auto-run JS ☒

Hello, cruel HTML & CSS world !

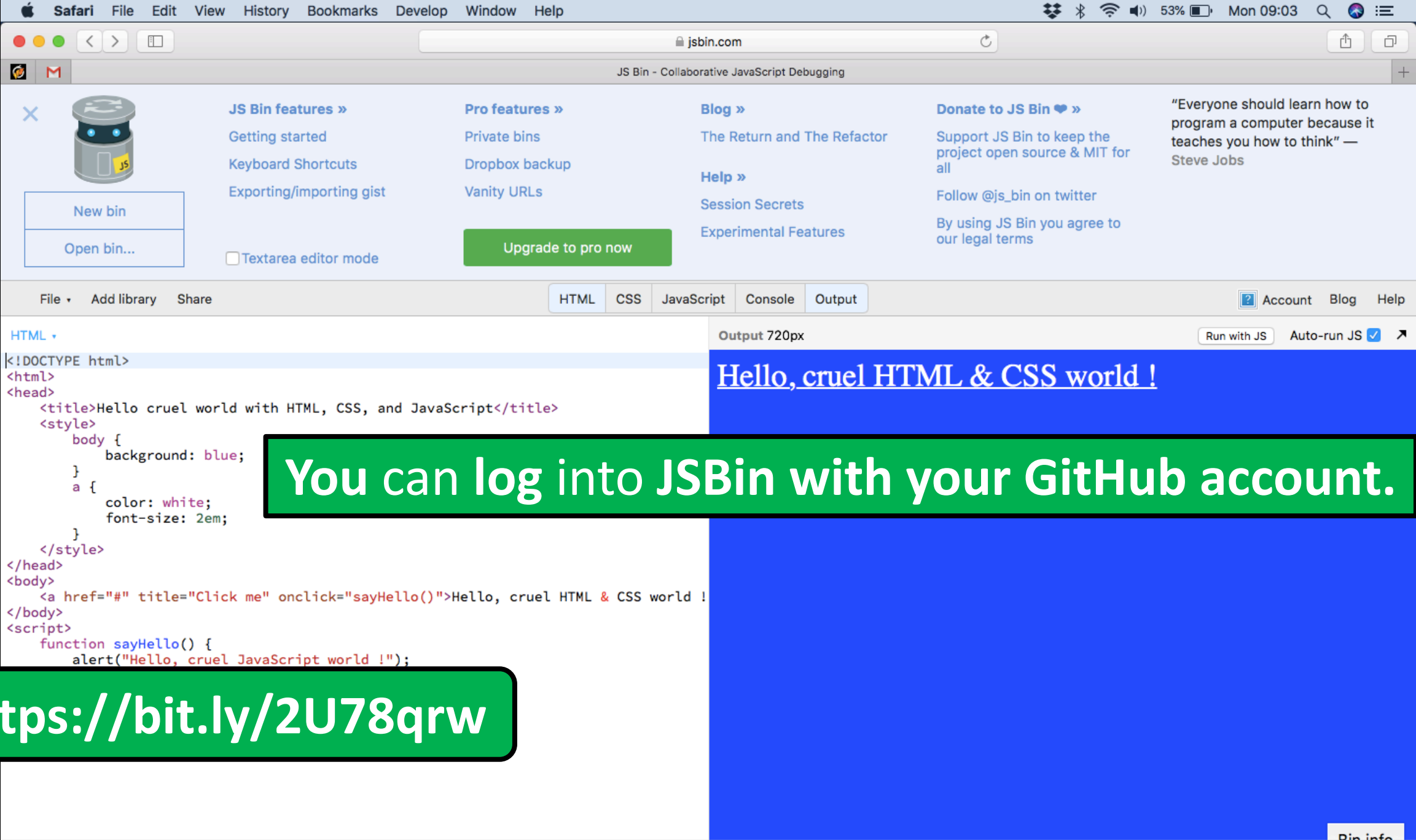
Hello, cruel JavaScript world !

Close

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You can log into JSBin with your GitHub account.

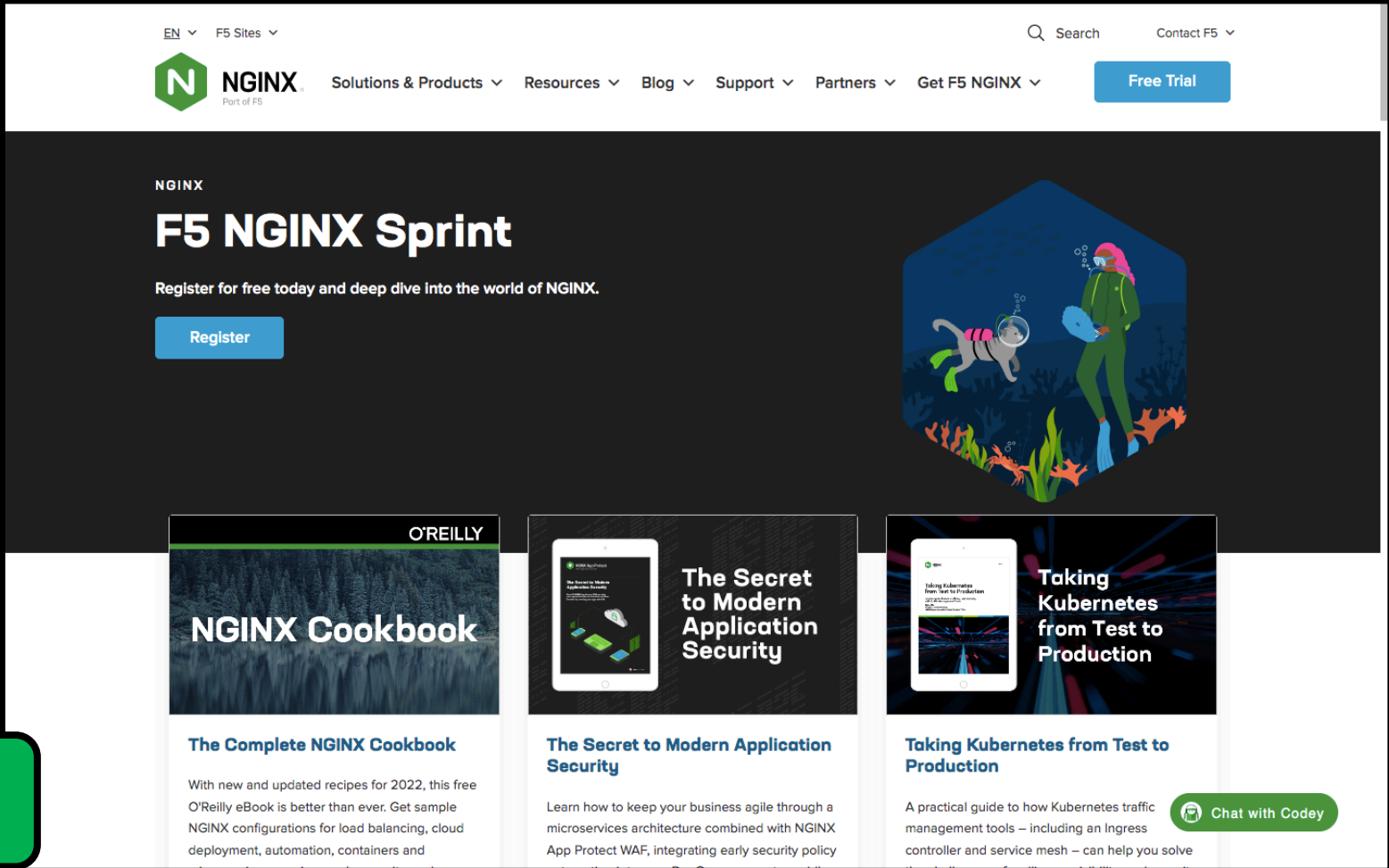
<https://bit.ly/2U78qrw>

Deploying and running the example in a local web server

Lecture: Deploying and running the example in a local web server

Module: Writing, deploying and accessing websites

There are several web servers out there.
We use **NGINX** in this example



The screenshot shows the NGINX website homepage. The header includes the NGINX logo (a green 'N' in a hexagon) and the text 'Part of F5'. Navigation links include 'Solutions & Products', 'Resources', 'Blog', 'Support', 'Partners', 'Get F5 NGINX', and a 'Free Trial' button. A search bar and 'Contact F5' link are also present. The main content area features a large banner for 'F5 NGINX Sprint' with the text 'Register for free today and deep dive into the world of NGINX.' and a 'Register' button. To the right of the banner is an illustration of a person and a dog diving underwater. Below the banner are three featured eBooks: 'NGINX Cookbook' by O'Reilly, 'The Secret to Modern Application Security', and 'Taking Kubernetes from Test to Production'. Each eBook card includes a thumbnail image, the title, and a brief description. A 'Chat with Codey' button is located in the bottom right corner.

EN F5 Sites

Search Contact F5

NGINX Part of F5

Solutions & Products Resources Blog Support Partners Get F5 NGINX Free Trial

NGINX

F5 NGINX Sprint

Register for free today and deep dive into the world of NGINX.

Register

O'REILLY

NGINX Cookbook

The Complete NGINX Cookbook

With new and updated recipes for 2022, this free O'Reilly eBook is better than ever. Get sample NGINX configurations for load balancing, cloud deployment, automation, containers and

The Secret to Modern Application Security

Learn how to keep your business agile through a microservices architecture combined with NGINX App Protect WAF, integrating early security policy

Taking Kubernetes from Test to Production

A practical guide to how Kubernetes traffic management tools – including an Ingress controller and service mesh – can help you solve

Chat with Codey

<https://www.nginx.com>

NGINX open-source can be found here.



NGINX Open Source

The open source web server that powers more than 400 million websites

Current version: 1.21.0 • [Documentation](#)



NGINX Plus

The only all-in-one load balancer, web server, content cache, and API gateway

Current version: R24 • [Documentation](#)



NGINX Controller

Centralized monitoring and management for NGINX Plus

Current version: 3.16.0 • [Documentation](#)



NGINX App Protect

Modern app security powered by F5 WAF technology running on NGINX Plus

Current version: 2.2 • [Documentation](#)



NGINX Unit

Multi-language application server from NGINX

Current version: 1.24.0 • [Documentation](#)



NGINX Amplify

Lightweight, cloud-based monitoring for NGINX and NGINX Plus

Current version: 1.01 • [Documentation](#)

NGINX open-source can be found here.

[Free Trial](#)

Deliver Modern Applications at Scale with F5 NGINX

Deploy app and APIs faster and with more confidence than ever before.



NGINX Open Source

The open source web server that powers more than 400 million websites.



NGINX Plus

The all-in-one load balancer, reverse proxy, web server, content cache, and API gateway.



NGINX Instance Manager

Configure, scale, and manage NGINX Open Source and NGINX Plus instances in your enterprise.



NGINX Controller

Lightning-fast application delivery and API management



NGINX App Protect

Modern WAF and Denial of Service to protect apps and



NGINX Ingress Controller

Kubernetes traffic management with API gateway, identity, and

[Chat with Codey](#)



Admin Guide

Installing NGINX and NGINX Plus

Installing NGINX Open Source

Install NGINX Open Source either as a prebuilt package or from source, following step-by-step instructions for all supported Linux distributions.

This article explains how to install NGINX Open Source.

What's on this Page

Choosing Between a Stable or a Mainline Version

Choosing Between a Prebuilt Package and Compiling from Source

We can compile NGINX from the source or use a prebuilt package. I prefer to use a package.

Google Cloud Platform

Installing NGINX Plus on Microsoft Azure

Deploying NGINX and NGINX Plus on Docker

> Basic Functionality

> Load Balancer

> Content Cache

> Web Server

- **Stable** – Doesn't include all of the latest features, but has critical bug fixes that are always backported to the mainline version. We recommend the stable version for production servers.

Choosing Between a Prebuilt Package and Compiling from Source

Both the NGINX Open Source mainline and stable versions can be installed in two ways:

- As a prebuilt binary package. This is a quick and easy way to install NGINX Open Source. The package includes almost all official NGINX modules and is available for most popular operating systems. See [Installing a Prebuilt Package](#).
- As binaries you compile from source. This way is more flexible: you can add particular modules, including third-party modules, or apply the latest security patches. See [Compiling and Installing from Source](#) for details.

Installing Prebuilt Ubuntu Packages

Installing SUSE Packages

Installing Prebuilt Alpine Linux Packages

Compiling and Installing from Source

Installing NGINX Dependencies

Downloading the Sources

Configuring the Build Options

Completing the Installation from Source

▼ Admin Guide

▼ Installing NGINX and NGINX Plus

Installing NGINX Plus

Installing NGINX Open Source

Installing NGINX Plus AMIs on

Installing a Prebuilt Package

Installing NGINX Open Source from a package is much easier and faster than building from source, but building from source enables you to compile in non-standard modules. Prebuilt packages are available for most popular Linux distributions, including CentOS, Debian, Red Hat Enterprise Linux (RHEL), SUSE Linux Enterprise Server (SLES), and Ubuntu. See [Linux packages](#) at **nginx.org** for the list of currently supported operating systems.

Modules Included in a Prebuilt Package

See [Configure Arguments](#) at **nginx.org** for the list of modules included in each prebuilt package.

Installing Prebuilt CentOS and RHEL Packages

What's on this Page

[Choosing Between a Stable or a Mainline Version](#)[Choosing Between a Prebuilt Package and Compiling from Source](#)

Installing a Prebuilt Package

[Modules Included in a Prebuilt Package](#)[Installing Prebuilt CentOS and RHEL Packages](#)

There are prebuilt NGINX packages for several GNU/Linux distributions: CentOS, Oracle Linux, Red Hat, Debian, Ubuntu and Suse. Follow the instructions on this page for the platform of your choice. I will demonstrate how to install on MacOS with HomeBrew (next slides).

> Web Server

> Security Controls

> Monitoring

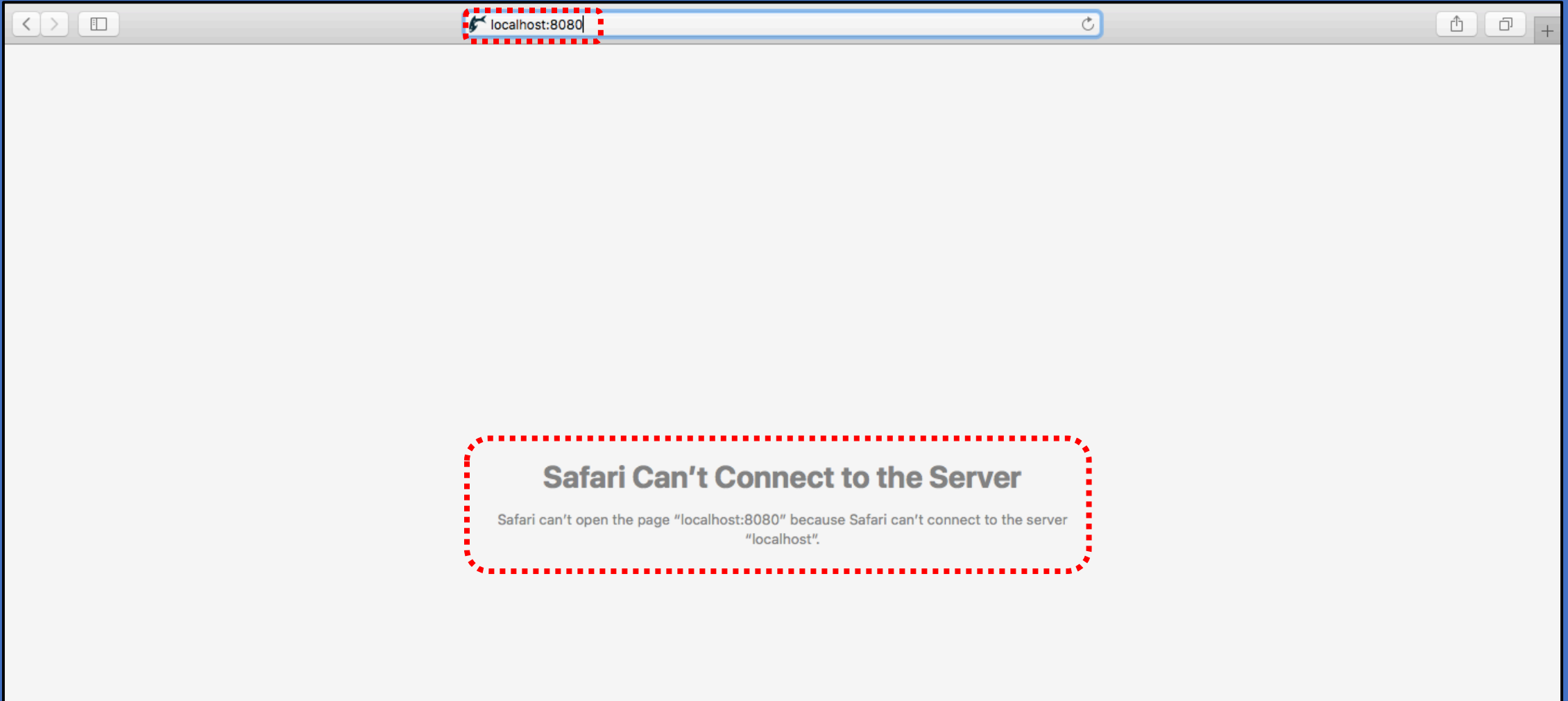
> High Availability

Installing a Prebuilt CentOS/RHEL Package from an OS Repository

1. Install the EPEL repository:

```
$ sudo yum install epel-release
```

2. Update the repository:



NGINX listens to port 8080. Let's first verify, before installing, that no other service is running in this port.

Terminal Shell Edit View Window Help
pagliares — -bash — 95x29
192:~ pagliares\$ brew install nginx

Installing Nginx on Mac OS using Homebrew

```
brew services start nginx
Or, if you don't want/need a background service you can just run:
nginx
==> Summary
🍺 /usr/local/Cellar/nginx/1.17.8: 25 files, 2.1MB
==> `brew cleanup` has not been run in 30 days, running now...
Removing: /Users/pagliares/Library/Caches/Homebrew/heroku--7.35.0.tar.xz... (6MB)
Removing: /Users/pagliares/Library/Caches/Homebrew/mysql--8.0.18_1.high_sierra.bottle.tar.gz... (76.7MB)
Removing: /Users/pagliares/Library/Logs/Homebrew/heroku... (114B)
Removing: /Users/pagliares/Library/Logs/Homebrew/mysql... (802B)
Removing: /Users/pagliares/Library/Logs/Homebrew/protobuf@3.7... (64B)
Removing: /Users/pagliares/Library/Logs/Homebrew/heroku-node... (119B)
Removing: /Users/pagliares/Library/Logs/Homebrew/openssl@1.1... (64B)
Pruned 10 symbolic links and 3 directories from /usr/local
```

Nginx can run as a background service or on demand.

The default port has been set in /usr/local/etc/nginx/nginx.conf to 8080 so that nginx can run without sudo.

nginx will load all files in /usr/local/etc/nginx/servers/.

To have launchd start nginx now and restart at login:

```
brew services start nginx
```

Or, if you don't want/need a background service you can just run:

```
nginx
```

```
192:~ pagliares$
```

Terminal Shell Edit View Window Help
pagliares — -bash — 96x29

```
192:~ pagliares$ nginx  
192:~ pagliares$  
192:~ pagliares$ netstat -ant | grep 8080  
tcp4          0      0  *.8080          *.*  
192:~ pagliares$
```

LISTEN

Starting Nginx and verifying it was started on port 8080 using the command line on Mac OS.



Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

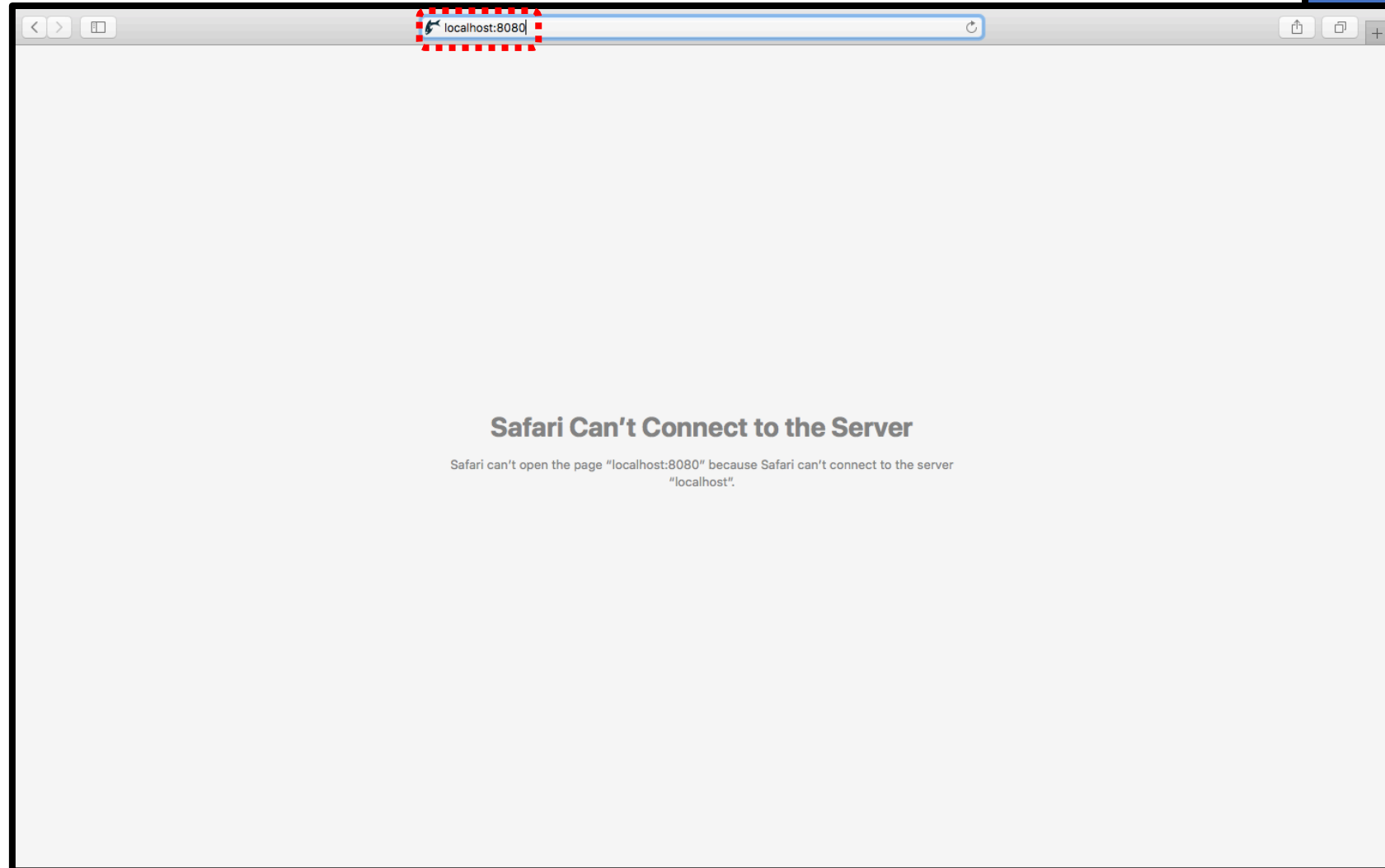
For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

Verifying whether NGINX was successfully started on port 8080 using a web browser.

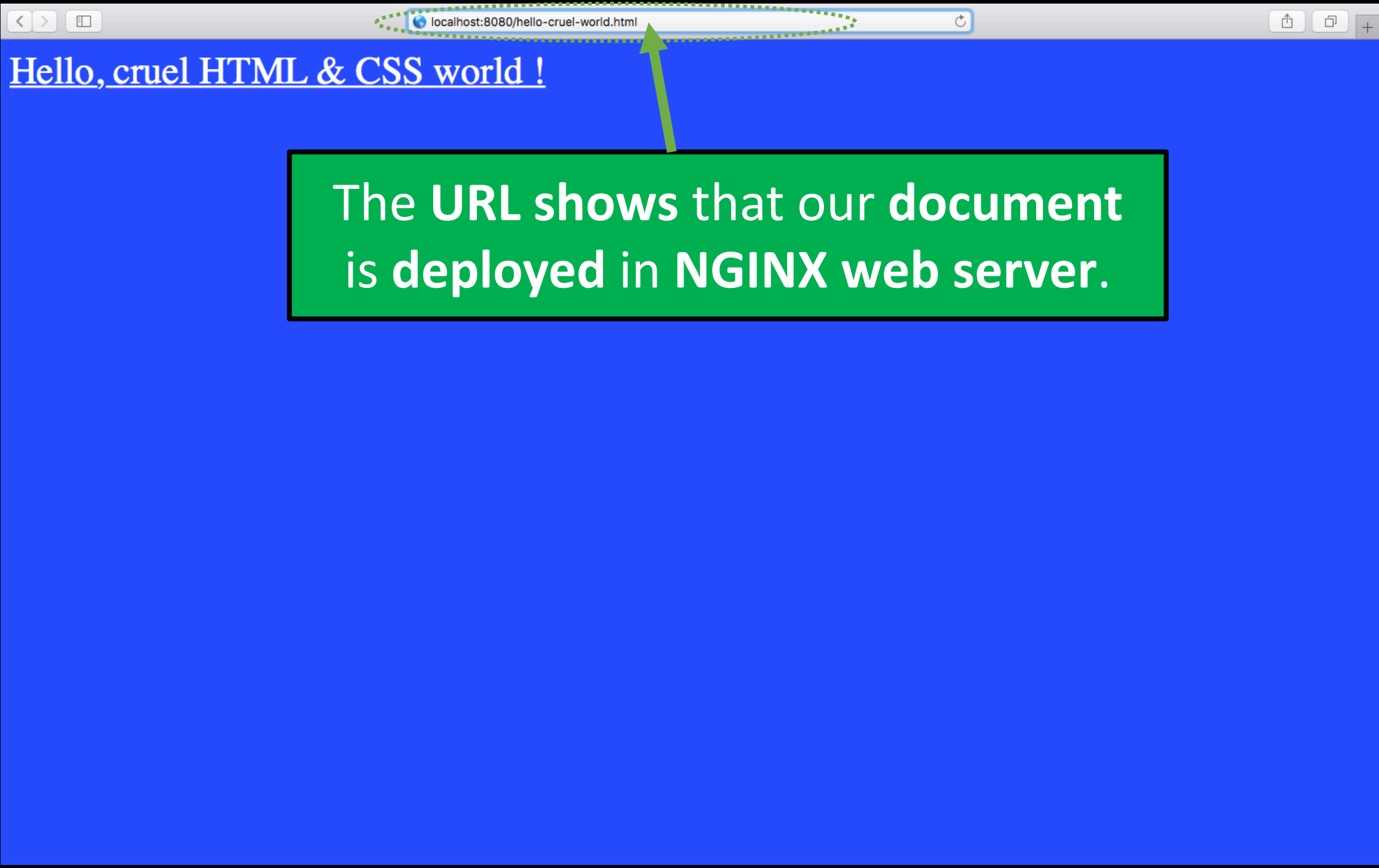
```
Rodrigos-MacBook-Air:~ pagliares$ nginx -s stop  
Rodrigos-MacBook-Air:~ pagliares$  
Rodrigos-MacBook-Air:~ pagliares$ netstat -ant | grep 8080  
Rodrigos-MacBook-Air:~ pagliares$
```

Stopping Nginx



```
192:Source_Code pagliares$ pwd
/Users/pagliares/Desktop/Source_Code
192:Source_Code pagliares$
192:Source_Code pagliares$ ls
hello-cruel-world.html
192:Source_Code pagliares$ cp hello-cruel-world.html /usr/local/var/www
192:Source_Code pagliares$
```

Copying (deploying) our web document to the appropriate directory in NGINX. Here, for simplicity, we deployed the web document in the root of the NGINX directory reserved for web applications. In a real scenario, we would have several distinct directories, each one holding a specific web application and we would deploy to some sub-directories of the directory representing the web application.



Hello, cruel HTML & CSS world !

The **URL** shows that our document
is **deployed** in **NGINX** web server.

Writing the example in Visual Studio Code with Live Server

Lecture: Writing the example in Visual Studio Code with Live Server
Module: Writing, deploying and accessing websites

**In this course, we are going to
use Visual Studio Code with Live
Server Extension**

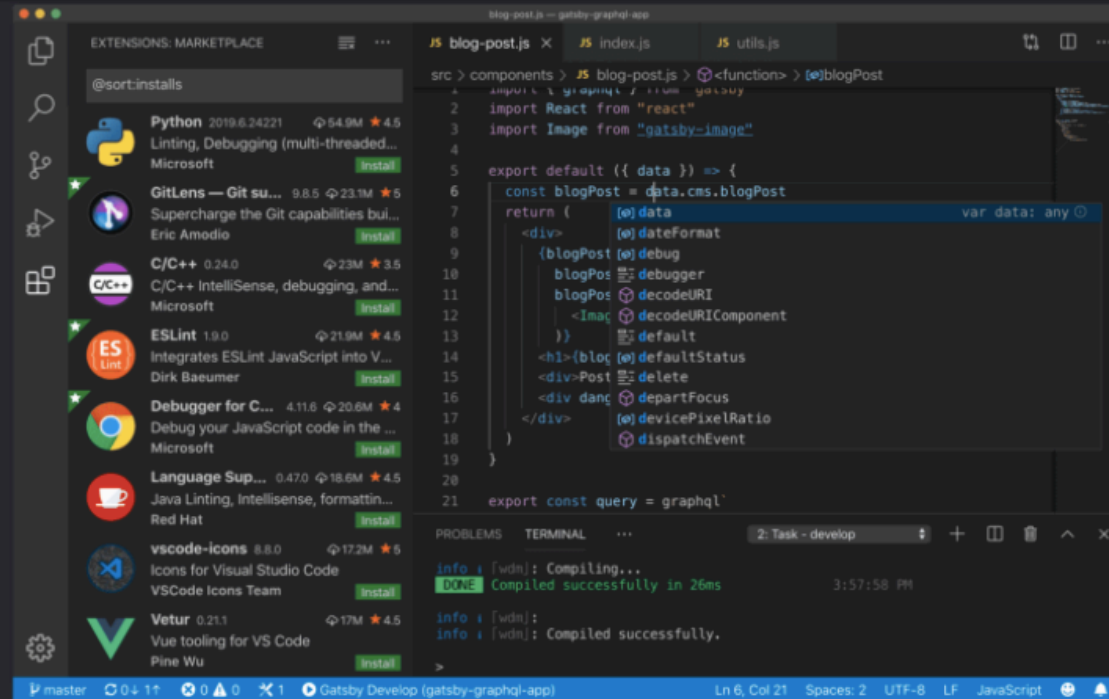
Code editing. Redefined.

Free. Built on open source. Runs everywhere.

Download Mac Universal
Stable Build

Other platforms and Insiders Edition

By using VS Code, you agree to its
license and privacy statement.



IntelliSense



Run and Debug



Built-in Git



Extensions

<https://code.visualstudio.com>

workspace / user preferences is

OS and use any VS Code plugins,

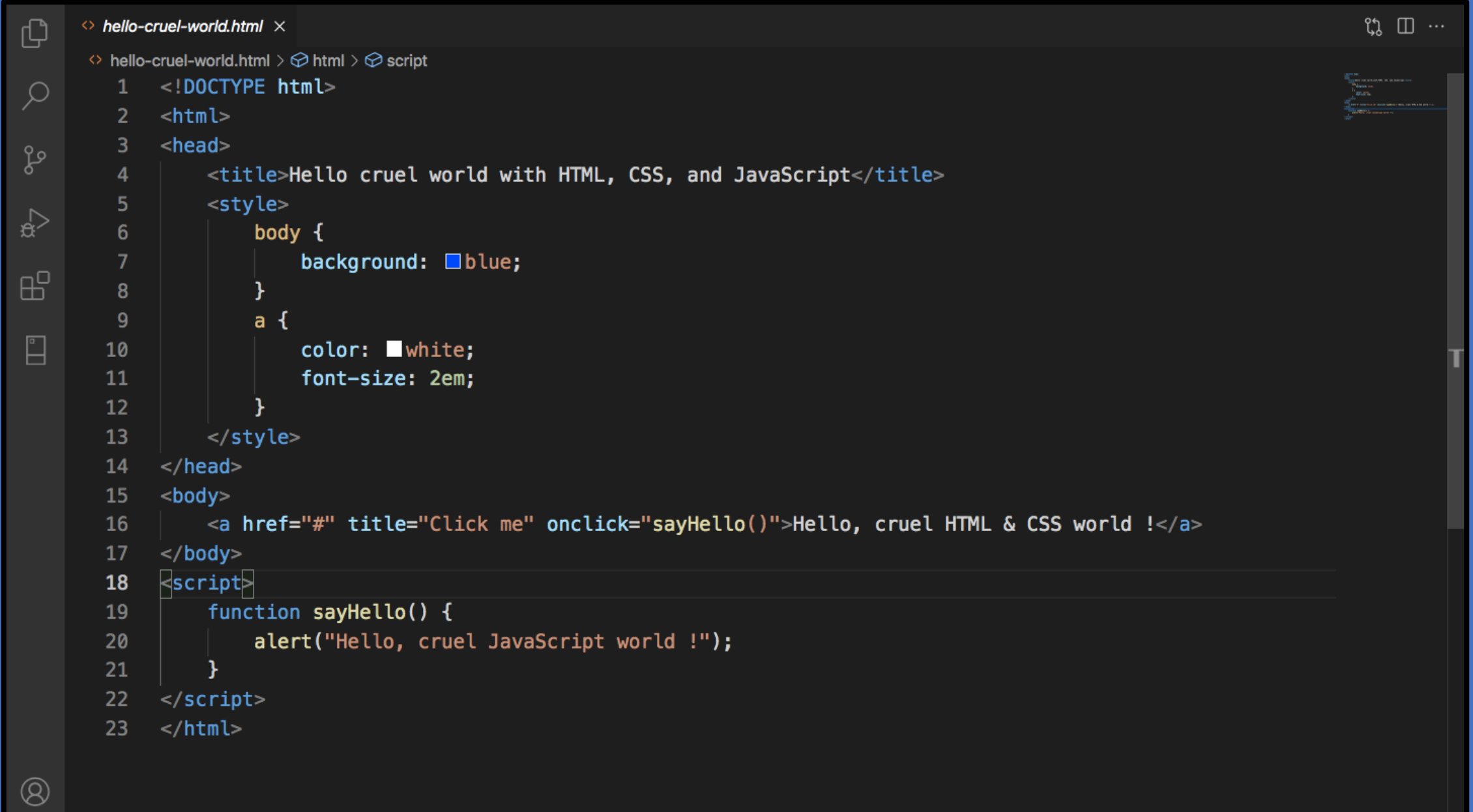
Dunlap @jadbox

will have deep remote
ment. You can connect to
er running a different



Pavithra Kodmad @PKodmad

VS Code is my most used and
favorite editor. I love being able
to customize the editor -
changing the layout, the icons,



The image shows a code editor window with a dark theme. The file name 'hello-cruel-world.html' is visible in the top-left corner. The breadcrumb navigation shows 'html > script'. The code is as follows:

```
1  <!DOCTYPE html>
2  <html>
3  <head>
4      <title>Hello cruel world with HTML, CSS, and JavaScript</title>
5      <style>
6          body {
7              background: blue;
8          }
9          a {
10             color: white;
11             font-size: 2em;
12         }
13     </style>
14 </head>
15 <body>
16     <a href="#" title="Click me" onclick="sayHello()">Hello, cruel HTML & CSS world !</a>
17 </body>
18 <script>
19     function sayHello() {
20         alert("Hello, cruel JavaScript world !");
21     }
22 </script>
23 </html>
```

web-programming-foundations/hello-cruel-world.html

Local web server

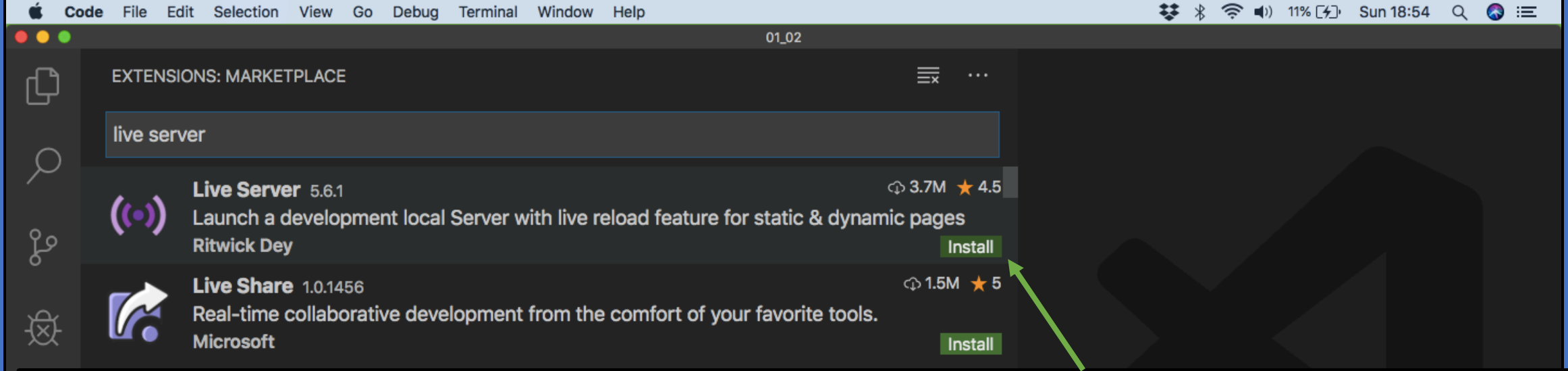
- As discussed earlier, to accurately replicate the web environment on our local computer, we need to run a local web server.
 - ✓ We used **NGINX** for this task.
- Other tools:
 - ✓ Vagrant
 - ✓ MAMP (comes bundled with **Apache** and **NGINX**)
 - ✓ Apache
 - ✓ ...
- But these are large, complex applications meant for large, complex sites that require set up and configuration.

Live servers

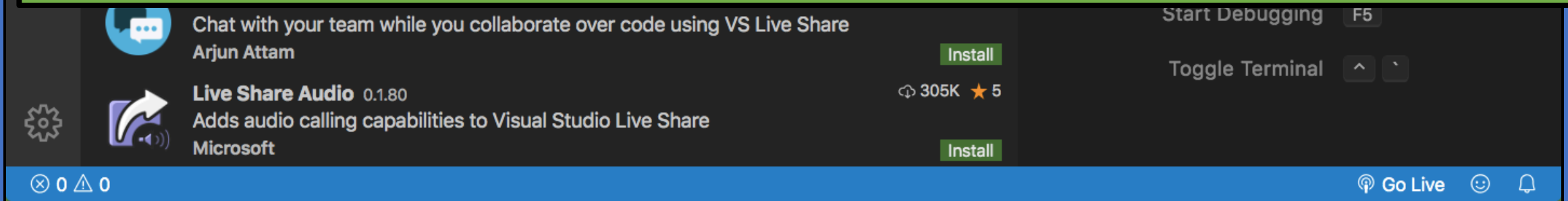
- If all we want to do is **quickly spin up** a couple of **HTML documents** with some **CSS** and **JavaScript**, there are **simpler ways** of **getting** the **real experience** in your **browser**.
- Most **modern code editors**, including **Visual Studio Code**, has **extensions** for **live servers**.
- These **extensions** add **functionality** to the **code editors**, to **spin up** a **small, local web server environment** that **syncs** to the **code editor** and the **browser**.

Live Server in Visual Studio Code

- This Live Server environment behave exactly as a live server somewhere else on the web, replicating the experience of publishing the site to the web.
- Anytime you make an edit to a file in the code editor, the server updates and the browser reloads the site on the fly.



In Visual Studio Code, you can use the extension Live Server, which adds a button to the bottom of your editing environment that says Go Live, and when you click on that button, a web server is spun up, and your browser is synced to the current live environment that's also synced to the editor.

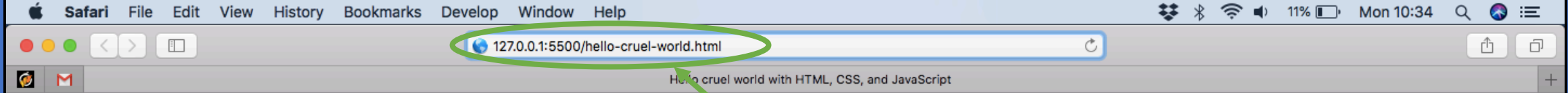


<> hello-cruel-world.html ×

Source_Code > <> hello-cruel-world.html > html

```
1  <!DOCTYPE html>
2  <html>
3  <head>
4      <title>Hello cruel world with HTML, CSS, and JavaScript</title>
5      <style>
6          body {
7              background: blue;
8          }
9          a {
10             color: white;
11             font-size: 2em;
12         }
13     </style>
14 </head>
15 <body>
16     <a href="#" title="Click me" onclick="sayHello()">Hello, cruel HTML & CSS world !</a>
17 </body>
18 <script>
19     function sayHello() {
20         alert("Hello, cruel JavaScript world !");
21     }
22 </script>
23 </html>
```

Go live !



We can verify our example running in live server through the address bar

Hello, cruel JavaScript world !

Close

Questions?